

Denver to Anchorage: A Detailed Optimization of Flight Variables

Rahul S. Gupta

Introduction

In the realm of aviation, there is a constant quest to achieve precision and efficiency. As modern aircraft technology advances and global air travel demands continue to rise, optimizing flight variables becomes increasingly important. The pursuit of optimal flight variables is driven by two necessities, safety and economics. The safety of passengers and crew is paramount in aviation, and calculations of aircraft performance are fundamental in ensuring the flight's safety and success. The other important factor in the pursuit of optimal flight variables is economic viability. In a world of finite resources, the management of fuel consumption along with numerous other variables directly impacts the flight's economic success.

Over the last couple of months, under the guidance of Dr. Ella Atkins, I have studied various aspects of flight for fixed-wing aircraft systems. I have been able to learn valuable fundamental principles in aircraft systems, manipulating and applying mathematical formulas to the various parts of an aircraft system. I have been able to apply my newfound mathematical skills in an engineering software known as MATLAB, which I learned under the guidance of Professor Atkins. With my knowledge of fixed-wing aircraft systems, I will be able to find steady flight variables for a twin jet engine aircraft and set up the necessary variables for a flight plan from Denver, Colorado to Anchorage, Alaska. First, we will dive into the parameters of the aircraft and how they are used. We will then discuss atmospheric pressure, temperature, and density, and how those variables will play a role in our flight. Once we have all of our parameters and our atmospheric variables, we will begin calculating variables that are required for steady-level flight, steady-level turning flight, climbing flight, and descending flight. All of the

calculations will be made through MATLAB, and all the code used in this paper can be found in the appendix. Through this paper, we will come to have a better understanding of the factors that come into play when determining an aircraft's flight plan.

Aircraft parameters

The first few variables we will discuss are weight variables, with max weight and fuel weight being given. This allows us to find the weight of the aircraft at different stages of flight, so for climbing flight, we will use max weight, for steady flight, we will use the difference between max weight and half the fuel weight, and for descending flight, we will use the difference between max weight and fuel weight (assuming that almost all of the fuel is used up by the time of descent). While weight is measured in pound-force (lbf), we must convert it to a force in Newtons (N) using the ratio of one lbf per 4.4482189 N. Another variable that is given in lbf is the max thrust which we will convert to N using the ratio above.

The next three variables are factors that describe the shape and geometry of the wings of the airplane. The first variable, planform area, is the total area of both wings from a top-down view. Planform area is measured in square meters and generally determines the lift and drag generated by the airplane. The next variable is span, also known as wingspan. Span is the distance from the tip of one wing to the other wing and is measured in meters. Lastly, the aspect ratio is the relationship between span and planform area. It is calculated by the quotient of the span squared and the planform area.

Dimensionless coefficients are values with no units that can be scaled up or down to accommodate better testing. There are numerous dimensionless coefficients in aviation, with aspect ratio being a great example. The aspect ratio allows a wing to be scaled down on that ratio

so that engineers can test its properties in wind tunnels. Along with aspect ratio, other given dimensionless coefficients are maximum lift coefficient, zero-alpha lift coefficient, lift slope, parasitic drag coefficient, and span efficiency factor.

These next few dimensionless coefficients will deal with the lift coefficient. The lift coefficient is the ratio of the aircraft's lift over the product of force times planform area. The maximum lift coefficient, which is given as 2.79 for our calculations, is the lift coefficient when the angle of attack (the angle between the aircraft's nose and the x-axis, also known as alpha) is greatest. The zero-alpha lift coefficient is the lift coefficient when the angle of attack is zero (the aircraft is directed parallel to the x-axis). Lastly, the lift slope represents the change in lift coefficient over the change in angle of attack. This means that it measures how much more lift can be generated by increasing the angle of attack.

Our final two dimensionless coefficients to discuss are the parasitic drag coefficient and span efficiency factor. The parasitic drag coefficient is a constant due to viscosity in an aircraft which is part of determining the overall drag coefficient. Along with parasitic drag, the other important factors in drag coefficient are lift coefficient, aspect ratio, and span efficiency. As we have already talked about lift coefficient and aspect ratio, we only have span efficiency to discuss. The span efficiency factor is a constant between zero and one that depends on the shape of the wing and evaluates the efficiency of the lift distribution along the wing's span. The closer the value of span efficiency is to one, the more effective the lift distribution is for the plane.

The values for all of the given variables discussed can be found in Table 1.

Table 1. *Aircraft Parameters for an Executive Jet*

Parameters	Symbol	Value
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Max Weight (full fuel)	W_{max}	489300 N
Fuel Weight	W_{fuel}	253550 N
Steady Flight Weight	W_{sf}	362530 N
Descent Weight (empty fuel)	W_{empty}	235760 N
Max Thrust	T_{ax}^{s}	30248 N
Planform Area	S	91.0449 m ²
Span (Wingspan)	b	28.956 m
Aspect Ratio	AR	9.2092
Max Lift Coefficient	C_{Lmax}	2.7900
Zero-alpha Lift Coefficient	C_{L0}	0.0190
Lift slope	$C_{L\alpha}$	0.1400
Parasitic Drag Coefficient	C_{D0}	0.0120
Span Efficiency Factor	e	0.9400

Atmospheric temperature, pressure, and density as functions of altitude

One of the most important factors when doing calculations related to aircraft flight is the atmospheric conditions. There are three ways to measure altitude in aerospace engineering: absolute, geometric, and true altitude. Absolute altitude is the height above the center of the Earth and is mainly used in space travel. True altitude, which is useful when flying very low to the ground, measures the height above a point on the surface of the earth and is only called upon for aircraft above mountain ranges or other tall structures. Geometric altitude is the most straightforward concept of altitude, representing how high an object is above the Earth's mean

sea level (MSL). Generally, geometric altitude is used when describing the altitude and will be used for this paper's calculations.

When it comes to aircraft, the three key factors that change as elevation increases are temperature, pressure, and density. Temperature is the average kinetic energy of a gas molecule and fluctuates at various intervals in our atmosphere. The levels of the atmosphere can be divided into two groups when discussing temperature. The first group is isothermal layers, which are atmosphere layers that maintain a constant temperature. The other group is constant-gradient layers, which change in temperature linearly with altitude at a specific lapse rate. The temperature and lapse rate at different points in the atmosphere can be seen below (Table 2).

Table 2. *Temperature Change through Atmospheric Layers*

Layer	Geometric Altitude (m)	Temperature (K)	Lapse Rate (K/m)
Troposphere	0	288.15	-0.0065
Tropopause	11000	216.65	0.0000
Stratosphere 1	20000	216.65	0.0010
Stratosphere 2	32000	228.65	0.0028
Stratopause	47000	270.65	0.0000
Mesosphere 1	51000	270.65	-0.0028
Mesosphere 2	71000	214.65	-0.0020
Mesopause	85000	186.65	0.0000

Atmospheric pressure is the force per unit area exerted by the weight of the air above a certain point. The weight of the air column is influenced by gravity and the density of the air. As

one moves higher in the atmosphere, the weight of the air decreases, resulting in lower atmospheric pressure. Pressure is measured in force per unit area (N/m²), also known as Pascals (Pa), and the standard atmospheric pressure at sea level is 101325 Pascals (Pa). Knowing the standard atmospheric pressure at sea level and the temperature at different levels in the atmosphere allows us to calculate the pressure at a specific height using two different equations, one pertaining to isothermal layers, and the other pertaining to constant-gradient layers. In isothermal layers, the equation for pressure change is given.

$$\frac{dp}{p} = - \left(\frac{1}{RT}\right)(g)(dh) \quad (1)$$

In this equation and many others used in physics, g is the gravitational constant (acceleration due to gravity on Earth) measured as 9.81 m/s², R is the gas constant for air measured as 287.053 J/(kg * k), and T is the temperature measured in K. By integrating the equation from the bottom of the layer to the top of the layer using h₁/p₁ as altitude/pressure at the bottom of the layer and h₂/p₂ as altitude/pressure at the top of the layer, we can simplify the equation to solve for p₂.

$$p_2 = (p_1)(e)^{-\left(\frac{g}{RT}\right)(h_2-h_1)} \quad (2)$$

The equation for constant gradient layers is given below.

$$\frac{dp}{p} = - \left(\frac{g}{R\xi}\right)\left(\frac{dT}{T}\right) \quad (3)$$

It is almost the same equation as the equation for isothermal layers, but the value of dh (change in height) is replaced by dT / T (change in temperature over temperature). Another change is the introduction of the symbol ξ , which represents the lapse rate. The integral of this equation from the bottom of the layer to the top of the layer using T₁/p₁ as temperature/pressure at the bottom of the layer and T₂/p₂ as temperature/pressure at the top of the layer gives us the equation for p₂.

$$p_2 = (p_1)\left(\frac{T_2}{T_1}\right)^{-\frac{g}{\xi R}} \quad (4)$$

With the equations for temperature and pressure at different altitudes, we are able to calculate the final and most important value for aviation, air density. The standard density is used in numerous calculations involving lift and drag, which means pretty much every calculation involving aircraft flight. Standard density, represented by the Greek letter rho (ρ), can be measured using the ideal gas law shown below.

$$p = \rho R T \quad (5)$$

Manipulating the variables allows us to put density in terms of temperature and pressure, as shown below.

$$\rho = \frac{p}{RT} \quad (6)$$

Thus, all of the calculations for temperature and pressure have culminated in us being able to find density at different points in the standard atmosphere, which, as said before, is extremely important for our calculations of steady flight, turning flight, climbing flight, and descent flight. The values for temperature (K), pressure (Pa), and density (kg / m³) from sea level to 100,000 meters can be seen in Figure 1.

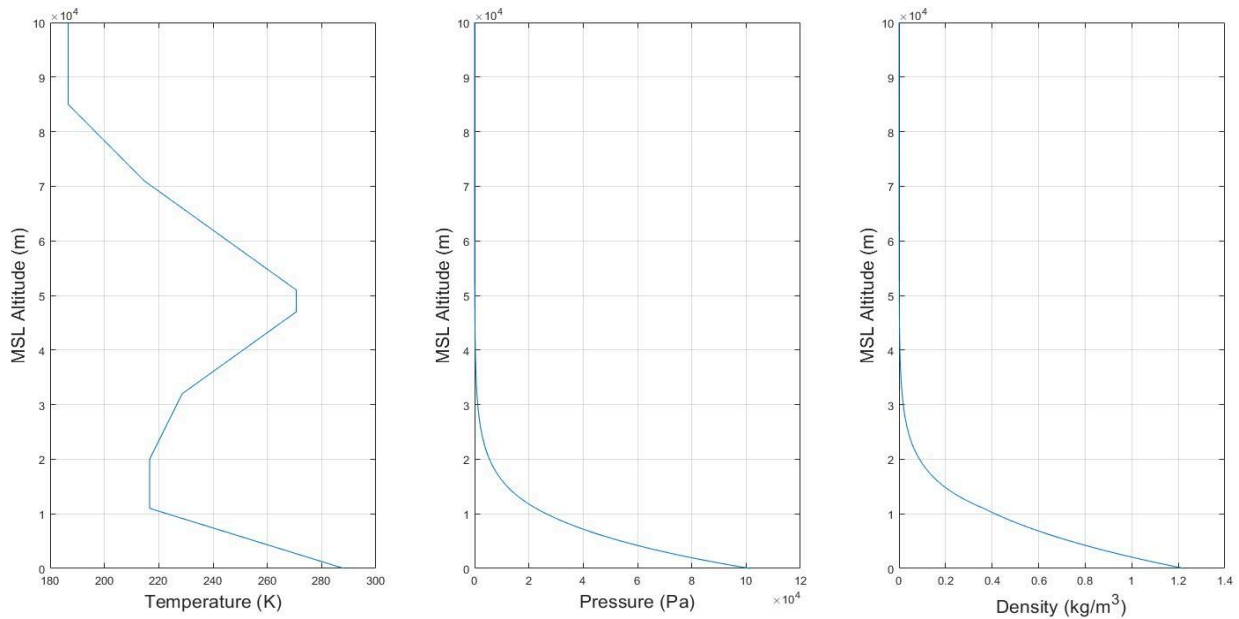


Figure 1. *Standard Atmospheric Temperature, Pressure, and Density*

As we can see, standard pressure and density follow a logarithmic pattern as altitude increases. This is due to the fact that at lower altitudes, the force of gravity compresses the air at a much higher rate than it does when altitude rises, so the air pressure and density are much higher at lower altitudes. As we look to optimize values such as airspeed, throttle, and angle of attack, it is important to recognize how temperature, pressure, and density play a significant role in our calculations.

Steady-level flight parameters

In steady-level flight for a twin-jet engine airplane, the most important variables are airspeed (velocity), thrust, throttle, and altitude. If we were discussing a propeller airplane, we would have to also account for power, which is the force that is created by a propeller instead of thrust, which is created by a jet engine, however, we will not discuss power for now. The first

variable to discuss is altitude, which we will set at one km or 10,000 meters for our calculations. 10,000 meters, which equates to about 32,000 feet, is the general cruising altitude for subsonic airplanes, as it is known to be most efficient in terms of drag due to the air density at 10,000 m. With altitude out of the way, we will move on to discuss the basics of steady-level flight.

Steady level flight means that all forces acting on the airplane are balanced, with lift equaling weight and thrust equaling drag, resulting in zero acceleration and a constant velocity. This is due to the equation $F = ma$, which means that force = mass x acceleration. Also known as Newton's Second Law, this basic principle of physics will allow us to substitute lift with weight and thrust with drag during our calculations.

The relationship between thrust and velocity which will allow us to calculate both stems from the equation for drag, given below.

$$D = \frac{1}{2} (\rho)(V^2)(S)(C_D) \quad (7)$$

In this equation, we already know density (ρ) and planform area (S). We still need a constant to replace C_D , which is the coefficient of drag, which is what the three equations below will help us do.

$$C_D = C_{D0} + \frac{(C_L)^2}{(\pi)(AR)(e)} \quad (8)$$

Now, we have parasitic drag (C_{D0}), aspect ratio (AR), and span efficiency factor (e), all known variables. We can find a value for the coefficient of lift through the next equation.

$$C_L = C_{L0} + (C_L\alpha)(\alpha) \quad (9)$$

Since the angle of attack (α) is equal to zero, the value for the coefficient of lift equals the value of the zero-alpha lift coefficient (C_{L0}), which is a known variable. Lastly, we will replace the known value for the coefficient of lift by putting it in terms of lift (L) using the equation below.

$$L = \frac{1}{2} (\rho)(V^2)(S)(C_L) \quad (10)$$

The final equation for drag in terms of lift is then shown below.

$$D = \frac{1}{2} (\rho)(V^2)(S)(C_{D0}) + \frac{2(L)^2}{(\rho)(V^2)(S)(\pi)(AR)(e)} \quad (11)$$

Since we are in steady-level flight, we can replace the drag force with thrust, and the lift force with weight, which turns the equation into a thrust equation with velocity unknown, shown below.

$$T = \frac{1}{2} (\rho)(V^2)(S)(C_{D0}) + \frac{2(W)^2}{(\rho)(V^2)(S)(\pi)(AR)(e)} \quad (12)$$

The resulting equation can create a graph with thrust as the y-axis and velocity as the x-axis, which I created below using the Desmos Graphing Calculator (Figure 2).

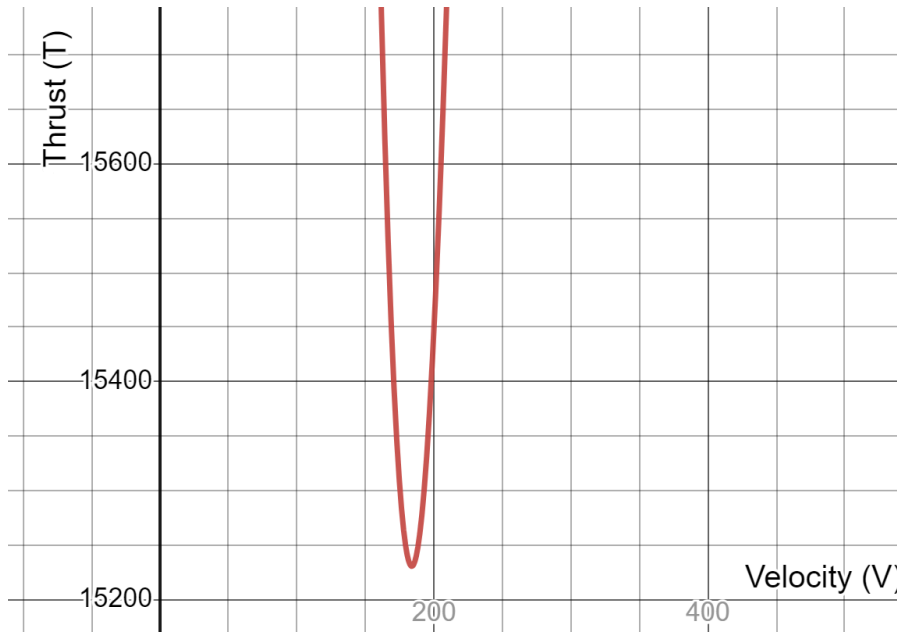


Figure 2. *Thrust (N) vs. Velocity (m/s) Graph*

The optimal velocity and thrust are found at the minimum on the velocity vs. thrust graph, which we can find by setting the derivative of the thrust equation with respect to velocity equal to zero. After taking the derivative of the thrust equation with respect to velocity, setting it equal to zero, and simplifying it to isolate velocity, we can find the minimum velocity equation, shown below.

$$V_{T_{i\ddot{}}}} = \left(\frac{4(W)^2}{(\rho)^2(S)^2(\pi)(AR)(e)(C_{D0})} \right)^{\frac{1}{4}} \quad (13)$$

Thus, we have put velocity into an equation in which we know all of the constants, giving us a steady-level flight velocity of 183.7885 m/s. Now that we know the velocity, we can plug it back into the thrust equation, giving us a thrust force of 15231 N. As seen in Figure 2, these values correspond to the coordinates at the minimum of the thrust vs. velocity graph.

Our final variable to calculate for steady-level flight is throttle. The throttle is a dimensionless value between zero and one which can be changed by the pilot to adjust airspeed and thrust. The equation for thrust in relation to the throttle is given below.

$$T = (\delta\ddot{}})(T^{s\ddot{}}_{ax})\left(\frac{\rho}{\rho_s}\right)^m \quad (14)$$

The equation can be simplified to isolate throttle ($\delta\ddot{}}$). The equation uses thrust (T), density (ρ), and density at sea level (ρ_s), all known values. The other values, $T^{s\ddot{}}_{ax}$ and m , must be changed for our calculations. The max thrust ($T^{s\ddot{}}_{ax}$) must be multiplied by two because the original max thrust is the value for one of the two jet engines. The value of m is a constant that changes based on the specific aircraft. For our aircraft, the value of m is 1/2. Knowing all of these constants, the value for throttle for our aircraft can be calculated as 0.4337, also known as 43.37% throttle.

Steady-turning flight parameters

In calculating steady-turning flight variables, the basic principles of balancing forces to create a constant velocity and zero acceleration still apply. The main difference comes in

calculating all of our variables to adjust for a higher necessary lift. The changes in thrust, drag, lift, and weight forces on the airplane can be seen in Figure 3.

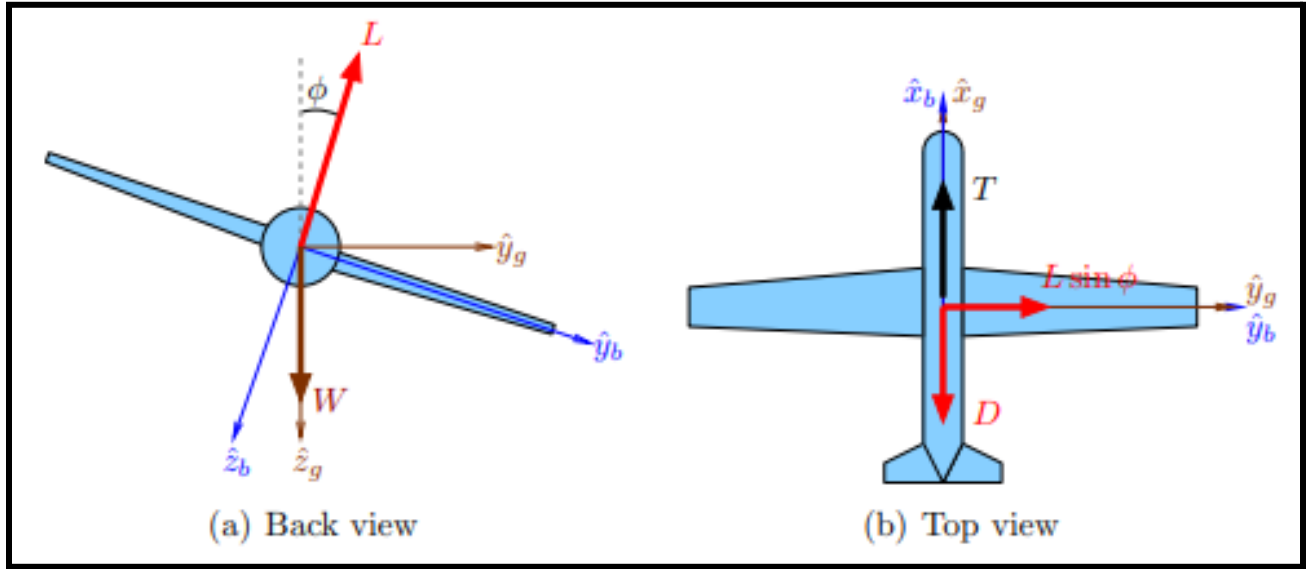


Figure 3. *Lift, Weight, Thrust, and Drag in steady turning flight* (Fidowski et al., 2019, p. 125)

The value for phi (ϕ) is known as the turn/bank angle, and for our calculations will be 25 degrees. Knowing the value for the bank angle, we are able to calculate lift using the equation below.

$$L \cos(\phi) = W \quad (15)$$

The value for lift is important because it allows us to calculate the value for load factor (n). Load factor is a dimensionless value that measures the ratio between lift and weight, and is normally one in steady-level flight, as lift and weight are equal in steady-level flight. The load factor is calculated using the equation below.

$$n = L/W = \frac{1}{\cos(\phi)} \quad (16)$$

Since we are calculating for steady turning flight, the values of lift (L) and weight (W) are not equal, and as a result, the load factor is greater than one. After calculating the value of lift during steady-turning flight, we are able to calculate the load factor during steady-turning flight, which is 1.1034 at a bank angle of 25 degrees. This is slightly more than the load factor at steady-level flight, meaning if you were sitting on the airplane, you would feel more than normal pressure on yourself when the airplane turns.

The next variable that is to be calculated is the turning radius, which we will use more physics principles to derive an equation for. As said before, $F = ma$, also known as Newton's second law, tells us that force = mass x acceleration. In centripetal motion, the F in $F = ma$ signifies the centripetal force, which is the force directed toward the center of the turning arc, which is calculated as $L\sin(\phi)$. Another important aspect of centripetal motion is that the acceleration in centripetal motion is equal to the difference between velocity squared and radius. The mass in this situation can be classified as W/g , with g being the force of gravity. By dividing by the weight value (W), which equals $L\cos(\phi)$, we can get $\tan(\phi)$ as the value on the left of the equation. As a result of all of this, we are left with the following equation.

$$\tan(\phi) = \frac{V^2}{gR} \quad (17)$$

We know the bank angle (ϕ), velocity (V) is the value calculated in steady-level flight, and the force of gravity (g) is 9.8 m/sec². Knowing all of these constants, we can find the turning radius (R) to be 739.16 meters.

The next couple of values we will calculate are thrust and throttle. All of them will be slightly more than they were for level flight, as we will need to generate more lift in turning flight. The equation for thrust in turning flight is shown below.

$$T = \frac{1}{2} (\rho)(V^2)(S)(C_{D0}) + \frac{2(W)^2}{(\rho)(V^2)(S)(\pi)(AR)(e)(\cos(\phi))^2} \quad (18)$$

The equation is nearly the same as that of thrust for level flight, with the only addition being the value for $(\cos(\phi))^2$ added. Since we are using the velocity from steady-level flight, we can simply plug in all of the known variables, which outputs a thrust value of 16,886 N of force. Using this thrust value, we can use the same equation we used for throttle in steady-level flight, finding our throttle in turning flight to be 0.4809, more easily referred to as 48.09% throttle. This value is about five percent greater than our throttle for steady-level flight.

Climbing and descending flight parameters

Let's move on to discuss climbing flight variables. In climbing flight, a new angle is added, the flight path angle (γ). The importance of flight path angle is that it affects the thrust required in the flight. The new value for thrust is shown below.

$$T = W\gamma + \frac{1}{2} (\rho)(V^2)(S)(C_{D0}) + \frac{2(W)^2}{(\rho)(V^2)(S)(\pi)(AR)(e)} \quad (19)$$

No longer can we just replace the D with its values, derive with respect to V, and set the derivative equal to zero. We must account for the unknown gamma (γ) value. One way to remove the gamma value from the equation is by using the equation below.

$$V_{climb} = V\gamma \quad (20)$$

As a result of this equation, we can replace the $W\gamma$ in the thrust equation resulting in a new equation.

$$T = W \frac{V_{climb}}{V} + \frac{1}{2} (\rho)(V^2)(S)(C_{D0}) + \frac{2(W)^2}{(\rho)(V^2)(S)(\pi)(AR)(e)} \quad (21)$$

By isolating the equation above so that V_{climb} is alone (the rate of climb), we are able to take the derivative of that equation and set it equal to zero to maximize V_{climb} . Since the new equation

still has a factor of thrust in it, we can make an assumption of operating at full throttle, resulting in the thrust being calculable with the equation.

$$T = (\delta t)(T^s \varnothing_{ax})\left(\frac{\rho}{\rho_s}\right)^m \quad (22)$$

Knowing all of the values in the thrust equation, and multiplying $T^s \varnothing_{ax}$ by two because of the two jet engines, we can find the climbing thrust. We can also find the V in equation 21 using equation 12, which is the equation that puts thrust as a function of velocity. Knowing the thrust in this equation, we can calculate V. Now that we know V, we can plug both velocity and thrust into the rate of climb equation shown below.

$$V_{climb} = V\left(\frac{T}{W}\right) - \frac{1}{2W}(\rho)(V^3)(S)(C_{D0}) - \frac{2W}{(\rho)(V)(S)(\pi)(AR)(e)} \quad (23)$$

After finding V_{climb} we know almost every variable to complete out climbing flight. We are able to calculate the value for our angle of attack (γ) using equation 20 and we will also be able to do basic estimates for our climbing flight, such as determining the total time for the climb, altitude at certain points during the climb, or distance traveled during the climb.

The final calculations we will have to make will be our descending flight calculations. The same principles that apply to climbing flight also apply to descending flight, with the only change required being a negative flight path angle (γ). While in climbing flight you can optimize for the flight path angle to be greatest, in descending flight, there is no optimization for the lowest possible flight path angle as it would result in an uncontrollable nosedive. Therefore, we can choose a safe flight path angle for descending flight and use equations 19-21 to calculate all of our descent variables, which we can use for estimates about total descent time, altitude at certain points during descent, or distance traveled during descent, ensuring a safe and efficient descent.

Conclusion

My experience learning from Dr. Atkins has given me valuable knowledge about aircraft systems that are heavily applicable to ensuring safety and efficiency in aviation. All of the steady flight, turning flight, climbing, and descending parameters calculated will allow for a flight plan to be created using waypoints from Denver to Anchorage. The calculations in this analysis of a twin-jet engine aircraft show how basic principles of physics and mathematical manipulations are used in the world of aviation. These calculations, used for all fixed-wing aircraft, allow for resource-efficient and safe travel. They save time, lower costs, reduce excess fuel usage, and lessen the environmental impact of the industry, all of which are essential resources for Earth and its people. In summary, efficient calculations in aviation are fundamental to the well-ordered functioning of the aviation industry and the well-being of all stakeholders involved.

References

Fidkowski, C., Atkins, E., & Powell, K. (2019). *Introduction to Aerospace Engineering*. Ann Arbor, MI: University of Michigan.

Appendix

Aircraft parameters

```
% All variables necessary

Wmax = 110000 * 4.4482189; % converting weight lbf to N
Wfuel = 57000 * 4.4482189; % converting weight lbf to N

WSF = (Wmax - (Wfuel/2)); % calculating weight at steady flight
Wempty = Wmax - Wfuel; % weight for descent calculations

Tsmax = 6800 * 4.44822; % max thrust converted from lbf to N

S = 980 * .092903; % planform area converted from ft^2 to m^2
b = 95 * .3048; % span converted from ft to m
AR = (b^2)/S; % aspect ratio

CLmax = 2.79; % max lift coefficient
CL0 = .019; % zero-alpha lift coefficient
CLa = .14; % lift slope
CDO = .012; % parasitic drag coefficient
e = .94; % span efficiency factor

AltDEN = 1656; % altitude at Denver Airport in m
AltANC = 46; % altitude at Anchorage Airport in m

AltSF = 10000; % altitude at steady flight in m
AltClimb = (AltDEN + AltSF)/2; % altitude for climb calculations in m
AltDesc = (AltANC + AltSF)/2; % altitude for descent calculations in m

TurnBankAngle = 25 * (pi / 180); % angle for banking flight converted from degrees to radians

[~, r0, ~] = StdAtmFunc(0); % density at sea level
[~, rClimb, ~] = StdAtmFunc(AltClimb); % density for climb calculations
[~, rDesc, ~] = StdAtmFunc(AltDesc); % density for descent calculations
[x, rSF, y] = StdAtmFunc(AltSF); % density for steady flight calculations
```

Atmospheric temperature, pressure, and density as functions of altitude

```
function [p, d, t] = StdAtmFunc(h)

g = 9.80665; % gravitational acceleration [m/s^2]
R = 287.053; % gas constant for air [J/kg.K]
p0 = 101325; % sea-level pressure [N/m^2]

if (h < 11000)
    t = 288.15 - (h * .0065);
else
    t = 216.65;
end

if (h < 11000)
    p = (p0)*((t/288.15)^(-g/(R*.0065)));
else
    p = (p0)*exp(-g*(h - 11000)/(R * t));
end

d = p / (R*t);
end
```

Steady-level flight and steady-turning flight parameters

```
% Steady Level Flight

% Minimum velocity for steady flight in m/s
VTmin = ((4 * (WSF)^2) / ((rSF)^2 * (S)^2 * (pi) * (AR) * (e) * (CD0)))^(1/4)

% Required thrust for steady flight in N
Tmin = (2 * WSF) * (sqrt((CD0) / ((pi) * (AR) * (e))))

% Required throttle for steady flight
ThrottleSF = (Tmin) / ((2 * Tmax) * ((rSF / r0)^(1/2)))

% Steady Level Turning Flight

% Lift in N
L = WSF / cos(TurnBankAngle);

% Load factor
n = L / WSF

% Required thrust for turning flight in N
Tturn = ((1/2) * (rSF) * (VTmin)^2 * (S) * (CD0)) + ((2 * (WSF)^(2)) / ((rSF) * (VTmin) ✓
^(2) * (S) * (pi) * (AR) * (e) * (cos(TurnBankAngle))^(2)))

% Required power for turning flight
Pturn = Tturn * VTmin

% Required throttle for turning flight
Throttleturn = (Tturn) / ((2 * Tmax) * ((rSF / r0)^(1/2)))

% Turning radius in m
TurnRad = (VTmin^2) / ((9.8) * tan(TurnBankAngle))
```